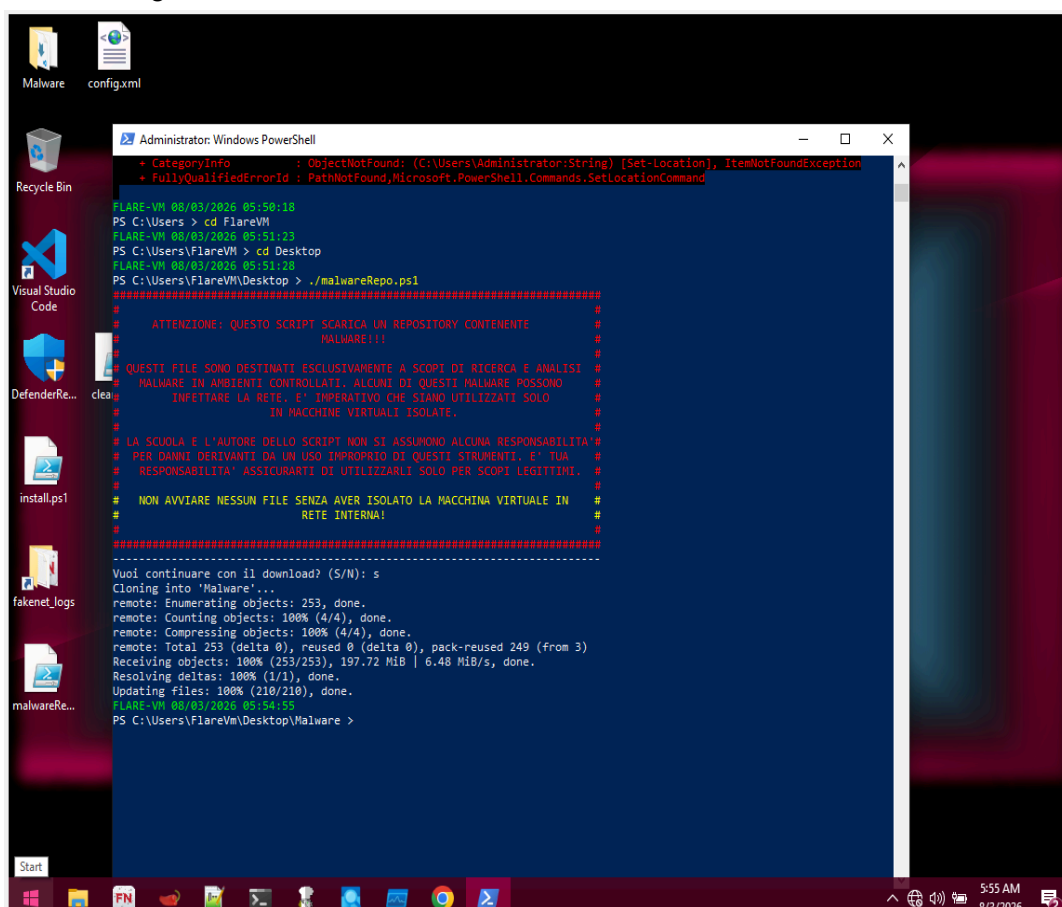


Relazione Tecnica: Simulazione di Persistenza e Analisi di Campioni di Malware

1. Introduzione e Scopo dell'Attività

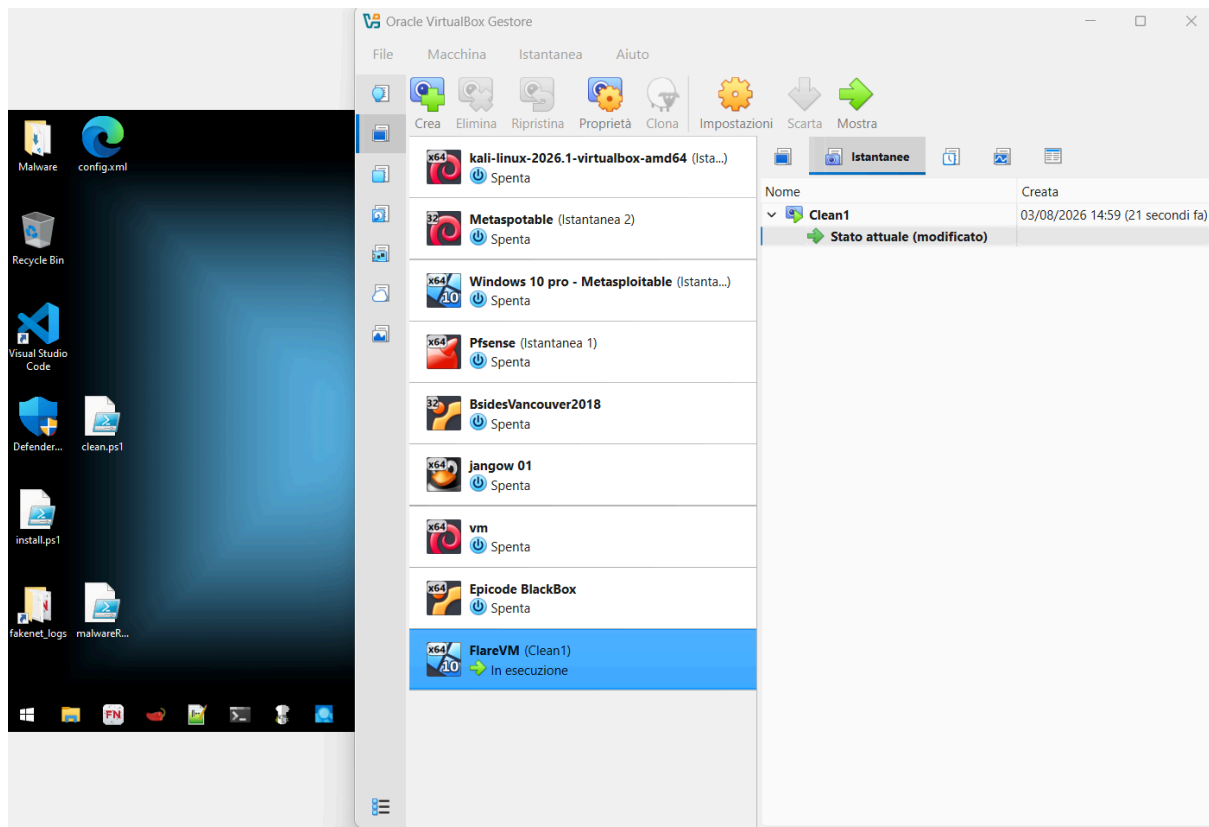
L'obiettivo di questa sessione di laboratorio è stato duplice:

- Simulare e comprendere le tecniche di **persistenza del registro di sistema** comunemente impiegate dai malware all'avvio del sistema operativo Windows.
- Estrarre, calcolare gli hash e analizzare in sicurezza **tre distinti campioni di software malevolo** utilizzando una macchina virtuale isolata (FlareVM) e servizi di threat intelligence come VirusTotal.



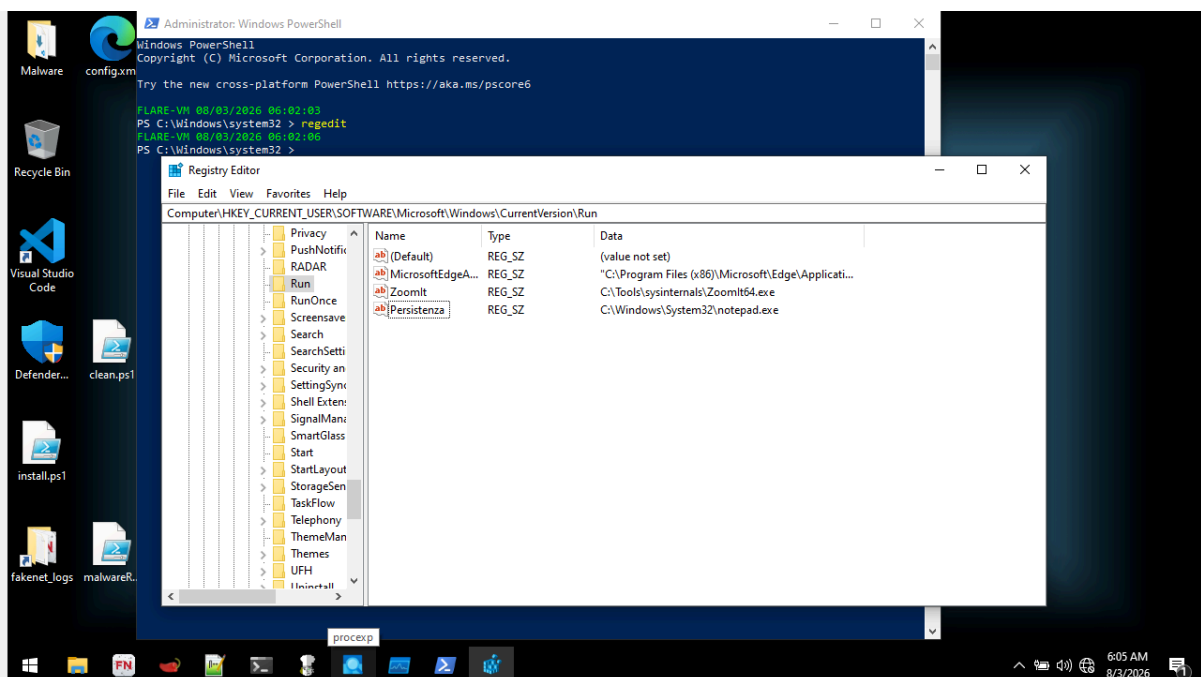
2. Preparazione dell'Ambiente e Messa in Sicurezza

Prima di procedere con l'estrazione e l'analisi dei file, l'ambiente di lavoro è stato messo in sicurezza:



- **Isolamento di rete:** La scheda di rete della macchina virtuale (FlareVM) è stata configurata in modalità di isolamento (Rete Interna) per prevenire comunicazioni accidentali o esfiltrazione di dati.
- **Snapshot pulita:** È stata creata un'istantanea di ripristino denominata **Clean1** su Oracle VirtualBox, utile per ripristinare lo stato iniziale pulito della VM.

3. Simulazione di Persistenza (HKCU Run)



Per comprendere come un malware mantenga l'accesso a un sistema dopo il riavvio, è stata configurata manualmente una chiave di persistenza nel Registro di Sistema:

- **Chiave modificata:**
HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run
- **Valore creato:** È stata inserita una stringa denominata **Persistenza** puntante all'eseguibile di sistema **C:\Windows\System32\notepad.exe**.
- **Verifica ed Effetto:** Al successivo riavvio della macchina virtuale, il sistema ha eseguito automaticamente il comando registrato, avviando il Blocco Note subito dopo il login dell'utente.

4. Analisi dei Campioni Malware e Calcolo degli Hash

Dalla cartella dei campioni sul Desktop, sono stati estratti e analizzati tre diversi file malevoli tramite il calcolo dei relativi hash (SHA-256):

1. **AgentTesla.exe** (Infostealer / Spyware)
2. **hbraeiwas.exe** (CrimsonRAT / Backdoor)
3. **metrofax.doc** (Documento Office con macro malevole / Dropper)

5. Analisi su VirusTotal e Tabella IOC (Indicator of Compromise)

L'interrogazione degli hash dei tre campioni sulla piattaforma VirusTotal ha permesso di delineare il profilo di rischio, esaminando la *Detection Ratio*, le relazioni di rete e i comportamenti in sandbox.

Campione / Nome file	Hash SHA-256	Rilevamento VT	Domini / URL C2 o IP associati	Descrizione / Famiglia
AgentTesla.exe	18aab0e981eee9e4ef8e15d4b003b14b3a1b0bfb7233fade8ee4b6a22a5abbb9	26 / 70	svc.ha-tea ms.office.com	Infostealer / Spyware (AgentTesla)

			104.86.18 2.8	
hbraei was.exe	dc31e710277eac1b125de6f4626765a2 684d992147691a33964e368e5f269cba	59 / 69	114.114.11 4.114 185.136.1 61.124	Trojan / Backd oor (Crim sonR AT)
metrof ax.doc	b13b29772c29ccb412d6ab360ff385258 36fcf0f65be637a7945a83a446dfd5e	46 / 62	bledroben. ru/gate.ph p dgfcomerc ial.com.br	Docu mento con macro / Dropp er (W97 M)

18aab0e981eee9e4ef8e15d4b003b14b3a1b0bfb7233fade8ee4b6a22a5abbb9

26
/ 70

Community Score -7

26/70 security vendors flagged this file as malicious

Reanalyze Similar More

18aab0e981eee9e4ef8e15d4b003b14b3a1b0bfb7233fade8ee4b6a22a5abbb9

AgentTesla.exe

Size 2.80 MB

Last Analysis Date 5 days ago

EXE

peexe detect-debug-environment nsis runtime-modules overlay direct-cpu-clock-access idle

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY 19

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat label hacktool.AgentTesla/docmo

Threat categories hacktool trojan dropper

Family labels agenttesla docmo negasteal

Security vendors' analysis

Do you want to automate checks?

AllCloud	HackTool:Win/BEL.Gen	ALYac	Application.HackTool.BEL
Antiy-AVL	Trojan/Win32.Agent	Arcabit	Application.HackTool.BEL
Arctic Wolf	Unsafe	BitDefender	Application.HackTool.BEL
CTX	Exe.trojan.agenttesla	eScan	Application.HackTool.BEL
Fortinet	Riskware/AgentTesla	GData	Application.HackTool.BEL
Google	Detected	Lionic	Hacktool.Win32.AgentTesla.3lc
MaxSecure	Trojan.Malware.338780217.susgen	McAfee Scanner	TII:18AAB0E981EE

18aab0e981eee9e4ef8e15d4b003b14b3a1b0bfb7233fade8ee4b6a22a5abbb9

18aab0e981eee9e4ef8e15d4b003b14b3a1b0bfb7233fade8ee4b6a22a5abbb9/relations

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Contacted Domains (9)

Domain	Detections	Created	Registrar
154.210.82.20.in-addr.arpa	0 / 91	-	-
245.50.188.52.in-addr.arpa	0 / 91	-	-
33.28.101.95.in-addr.arpa	0 / 91	-	-
a1672.dscr.akama1.net	0 / 91	1999-03-03	MarkMonitor Inc.
res.public.onecdn.static.microsoft	0 / 91	2023-05-05	MarkMonitor Inc.
svc.ha-teams.office.com	0 / 91	1999-04-20	MarkMonitor Inc.
svc.ms-acdc-teams.office.com	0 / 91	1999-04-20	MarkMonitor Inc.
windows.msn.com-ion.edgesuite.net	0 / 91	2001-04-02	MarkMonitor Inc.
www.microsoft.com	0 / 91	1991-05-02	-

Contacted IP addresses (45)

IP	Detections	Autonomous System	Country
104.86.182.8	1 / 91	20940	US
131.253.33.203	0 / 91	8068	US
151.101.22.172	0 / 91	54113	US
152.195.19.97	0 / 91	701	US
192.168.0.17	0 / 91	-	-
192.168.0.18	0 / 91	-	-
192.168.0.27	0 / 91	-	-
192.168.0.3	0 / 91	-	-
192.168.0.32	0 / 91	-	-
192.168.0.5	0 / 91	-	-

18aab0e981eee9e4ef8e15d4b003b14b3a1b0bfb7233fade8ee4b6a22a5abb9

AgentTesla.exe

Size2.80 MB

Last Analysis Date5 days ago

EXE

Community Score

peexe detect-debug-environment nsis runtime-modules overlay direct-cpu-clock-access idle

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY 39+

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Display grouped sandbox reports

CyberArk

0

0

0

0

0

0

0

0

0

CAPE Sandbox

0

1

0

0

0

10

2

0

0

QAnXin RedDrip

0

0

0

0

0

10

0

0

0

VenusEye Sandbox

0

0

0

0

0

8

0

0

0

VirusTotal Jujubox

0

0

0

0

0

0

0

0

0

CyberArk

0

8

0

0

0

0

0

0

0

Microsoft Sysinternals

0

0

0

0

0

99+

45

0

0

Tencent HABO

0

0

0

0

0

0

0

0

0

VirusTotal CuckooFork

0

0

0

0

0

0

0

0

0

Zenbox

0

3

0

0

0

80

8

0

0

Activity Summary

Download Artifacts

Full Reports

Help

Detections

Mitre Signatures

IDS Rules

Sigma Rules

Dropped Files

Network comms

NOT FOUND

1 LOW 41 INFO

NOT FOUND

NOT FOUND

1 PE_EXE 1 XML 286 OTHER

9 DNS 43 IP 1 JA3

dc31e710277eac1b125de6f4626765a2684d992147691a33964e368e5f269cba

hbraeiwas.exe

Size84.00 KB

Last Analysis Date20 days ago

EXE

Community Score

peexe detect-debug-environment runtime-modules assembly executes-dropped-file long-sleeps service-scan spreader direct-cpu-clock-access

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY 18+

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat label

trojan.msil/crimsonrat

Threat categories

trojan ransomware dropper

Family labels

msil crimsonrat foreign

Security vendors' analysis

Do you want to automate checks?

AhnLab-V3	Trojan/Win.AN.C5234207	Alibaba	Backdoor:MSIL/CrimsonRat.88ea0f22
AliCloud	Backdoor:MSIL/CrimsonRat.A90KG	ALYac	IL-Trojan.MSIL.Zilla.9959
Antiy-AVL	Trojan/APT/MSIL.APT36	Arcabit	IL-Trojan.MSIL.Zilla.D26E7
Arctic Wolf	Unsafe	Avast	MSIL:CrimsonRAT-J [Rat]
AVG	MSIL:CrimsonRAT-J [Rat]	Avira (no cloud)	TR/Dropper.MSIL.Gen8
BitDefender	IL-Trojan.MSIL.Zilla.9959	Bkav Pro	W32.Malware.365F93D
ClamAV	Win.Trojan.CrimsonRAT-7591455-0	CrowdStrike Falcon	Win/malicious_confidence_100% (W)

BenveEPICO(S) DiOne PAlextoCS032LeetCHackeDNSOnlinifrostLinuxCentriChiarivxDocuNuovo Chrome disponibile

virustotal.com/gui/file/dc31e710277eac1b125de6f4626765a2684d992147691a33964e368e5f269cba/relations

dc31e710277eac1b125de6f4626765a2684d992147691a33964e368e5f269cba

DETECTIONDETAILSRELATIONSBEHAVIORCOMMUNITY

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Contacted Domains (4)

Domain	Detections	Created	Registrar
a1672.dscr.akamai.net	0 / 91	1999-03-03	MarkMonitor Inc.
elp-terr-na.cdp1.digicert.com.akahost.net	0 / 91	2015-12-04	-
res.public.onecdn.static.microsoft	0 / 91	2023-05-05	MarkMonitor Inc.
www.microsoft.com	0 / 91	1991-05-02	-

Contacted IP addresses (29)

IP	Detections	Autonomous System	Country
104.71.214.69	0 / 91	20940	US
104.86.182.10	0 / 91	20940	US
114.114.114.114	3 / 91	21859	CN
151.101.22.172	0 / 91	54113	US
184.27.218.92	0 / 91	20940	US
185.136.161.124	4 / 91	29066	FR
192.168.0.108	0 / 91	-	-
192.168.0.27	0 / 91	-	-
192.168.0.62	0 / 91	-	-
192.229.211.108	0 / 91	-	US

Execution Parents (52)

Scanned	Detections	Type	Name
2025-08-17	3 / 61	7ZIP	Toxic.7z
2025-09-25	2 / 65	ZIP	CrimsonRAT.zip

30°C Soleggiato

Cerca

15:18 03/08/2026

BenveEPICO(S) DiOne PAlextoCS032LeetCHackeDNSOnlinifrostLinuxCentriChiarivxDocuNuovo Chrome disponibile

virustotal.com/gui/file/dc31e710277eac1b125de6f4626765a2684d992147691a33964e368e5f269cba/behavior

dc31e710277eac1b125de6f4626765a2684d992147691a33964e368e5f269cba

DETECTIONDETAILSRELATIONSBEHAVIORCOMMUNITY

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

☒ Display grouped sandbox reports

☒ C2AE	0 0 0 0 0 1 1	☒ CAPA	0 1 0 0 0 0 0
☒ CAPE Sandbox	0 2 0 0 0 0 1	☒ DAS-Security Orcas	1 4 0 0 0 0 2
☒ DrWeb vxCube	1 0 1 2 3 0	☒ Microsoft Sysinternals	0 0 0 0 0 95 23
☒ QianXin RedDrip	0 0 0 0 0 2 0	☒ Sangfor ZSand	0 0 0 0 0 2 2
☒ VirusTotal Jujubox	0 0 0 0 0 0 1	☒ Zenbox	4 6 0 1 12 6

Activity Summary

Download ArtifactsFull ReportsHelp

4 Detections

3 MALWARE 1 TROJAN 1 EVADER 1 RAT

Mitre Signatures

1 LOW 23 INFO

IDS Rules

1 MEDIUM

Sigma Rules

1 HIGH 2 LOW

Dropped Files

97 OTHER 1 ZIP 1 PE_EXE 1 TEXT

Network comms

4 DNS 26 IP

Behavior Tags

checks hostname detect debug environment direct cpu clock access executes dropped file long sleeps runtime modules service scan

30°C Soleggiato

Cerca

15:19 03/08/2026

BenveEPICO(S) DkOne PAlexToCS032LeetCHackeDNSOnlinifrostLinuxCentriChiarivDocu+Nuovo Chrome disponibile

virustotal.com/gui/file/b13b29772c29ccb412d6ab360ff38525836cf0f65be637a7945a83a446dfd5e

b13b29772c29ccb412d6ab360ff38525836cf0f65be637a7945a83a446dfd5e

46/62

Community Score

-194

46/62 security vendors flagged this file as malicious

ReanalyzeSimilarMore

b13b29772c29ccb412d6ab360ff38525836cf0f65be637a7945a83a446dfd5e

Size221.50 KB

Last Analysis Date5 months ago

DOC

doc

checks-user-input

obfuscated

open-file

macro-run-file

detect-debug-environment

auto-open

create-ole

run-file

macros

calls-wmi

long-sleeps

malware

attachment

exe-pattern

environ

DETECTIONDETAILSRELATIONSBEHAVIORCOMMUNITY20+

Join our Community

and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Code insights

The macro defines several subroutines and functions. Auto_Open, AutoOpen, and Workbook_Open are auto-execution entry points, ensuring the macro runs when the document is opened. Mansdsa simply calls NuFaS3ssa. NuFaS3ssa constructs several strings. It extracts the year ("7037") from a date string, takes the first digit ("7"), and subtracts 8, resulting in the value -1. It uses this value in character code calculations and concatenations to build file paths and extensions. Environment variables are used in the creation of these paths. Specifically, the "TMP" environment variable is accessed and appended with ".ld". Strings ".exe" and ".rtf" are constructed using character codes.

Show more

Crowdsourced AI

Hispacec flags this file as malicious

The macros extracted from the document exhibit several signs of malicious intent.

Show more

Popular threat label

dropper.w97m/adnel

Threat categories

dropper

trojan

downloader

Family labels

w97m

adnel

stbij

30°C Soleggiato

Cerca

15:20 03/08/2026

The first screenshot shows the VirusTotal interface for the file `b13b29772c29ccb412d6ab360ff38525836cf0f65be637a7945a83a446dfd5e` (metrofax.doc). The file is 221.50 KB and was last analyzed 5 months ago. It has a Community Score of 46/62 and is flagged as malicious by 46/62 security vendors. The interface shows tabs for DETECTION, DETAILS, RELATIONS, BEHAVIOR, and COMMUNITY. The RELATIONS tab is active, displaying a table of contacted URLs and domains.

Scanned	Detections	Status	URL
2025-08-18	13 / 97	-	http://billedroben.ru/gate.php
2025-08-18	2 / 97	404	http://dgcmercial.com.br/wp-content/plugins/prism-highlight/opera1.exe
2025-08-18	11 / 97	-	http://rohironrof.ru/gate.php
2025-08-18	2 / 97	-	http://eloraestate.com/wp-content/plugins/prism-highlight/opera1.exe
2025-08-18	2 / 97	-	http://edmontonlimo247.com/wp-content/plugins/prism-highlight/opera1.exe
2025-08-18	11 / 97	-	http://fouseevenghedt.ru/gate.php

Domain	Detections	Created	Registrar
billedroben.ru	13 / 91	-	-
dgcmercial.com.br	0 / 91	-	-
edmontonlimo247.com	1 / 91	2013-10-23	GoDaddy.com, LLC

The second screenshot shows the same file analysis but with the BEHAVIOR tab active. It displays a table of sandbox reports from various vendors, including C2AE, DOCGuard, VMRay, CAPE Sandbox, Tencent HABO, and Zenbox. The table shows the number of detections and the status of the file in each sandbox.

Sandbox	Detections	Status
C2AE	12	Malicious
DOCGuard	2	Malicious
VMRay	1	Malicious
CAPE Sandbox	2	Malicious
Tencent HABO	0	Not analyzed
Zenbox	5	Malicious

6. Conclusioni

L'esercitazione ha permesso di completare con successo l'analisi di tre differenti vettori d'attacco (un eseguibile infostealer, un RAT e un documento di Office armato con macro). Attraverso l'uso di ambienti virtualizzati isolati, strumenti di hashing e piattaforme di threat intelligence come VirusTotal, è stato possibile tracciare con precisione gli indicatori di

compromesso (IOC), confermando l'efficacia delle metodologie di analisi forense e malware analysis di base.